

MARIAN COLLEGE KUTTIKANAM (AUTONOMOUS)
UG Programme (Honors) Degree March 2026 Regular Examinations

Second Semester

252CCRUBC102 Data Structures

Time:

Maximum Marks: **60**

Answer one question from each bunch

BUNCH I

- I.A ✓ While developing an online shopping application, Kiran is trying to search for a specific product from a list of items stored in the system. To make the search faster, he decides to use a binary search algorithm. However, he notices that the list of products is not sorted in any particular order. Does this situation create a problem for using the binary search algorithm? Explain why or why not. Also, write the algorithm for binary search and demonstrate the searching process with a suitable example. (CO1:15 Marks)

OR

- I.B In a digital library catalog system, a list of book IDs has been sorted in ascending order as follows: [1, 8, 12, 34, 45, 57, 77]. The system needs to locate the book with ID 77 from this list. Two search methods available are: i) Linear Search ii) Binary Search. Based on your understanding, identify which algorithm would locate the element 77 more efficiently, and justify your choice. Also, demonstrate how the selected algorithm works step by step on the given list, and provide the algorithm used, explaining why it is more suitable for searching in this case. (CO1:15 Marks)

BUNCH II

- II.A.1 A software developer is simulating function call tracking in a program using a stack data structure. The following sequence of operations is performed on the stack: PUSH(10), PUSH(20), POP, PUSH(10), PUSH(20), POP, POP, POP, PUSH(20), POP. Demonstrate the state of the stack after each operation and explain how the stack handles the

sequence, especially when multiple POP operations are executed in succession. (CO2:10 Marks)

II.A.2 Meera is working on a compiler design project where arithmetic expressions must be converted into a format that can be easily evaluated by the system. To achieve this, the system converts infix expressions into prefix form. She is given the following infix expression: $(A + B) * (C - D) / E$

Convert the given infix expression into its equivalent prefix form, clearly showing each step involved in the transformation process. (CO2:5 Marks)

OR

II.B In a warehouse dispatch system, a **linear queue Q** with a maximum size of 5 is used to manage boxes waiting to be loaded onto delivery trucks. The system currently holds the following boxes in the queue: 100, 504, 353, 475, 756. Demonstrate how the linear queue handles the **deletion** of each box step by step as they are loaded. Provide the **algorithm used for deleting elements** from a linear queue and also show what happens when a **deletion is attempted after the queue becomes empty**, illustrating the **underflow condition**.

(CO2:15 Marks)

BUNCH III

III.A.1 You are developing a **contact management system** for a mobile phone. The system needs to store a **list of contacts**, which undergoes **frequent additions and deletions** as users regularly update, add, or remove entries. The system must also allow **quick access to any contact by index**. As the lead developer, you must choose the most appropriate data structure, **Array or Linked List** to implement the contact list. Compare and contrast Arrays and Linked Lists.

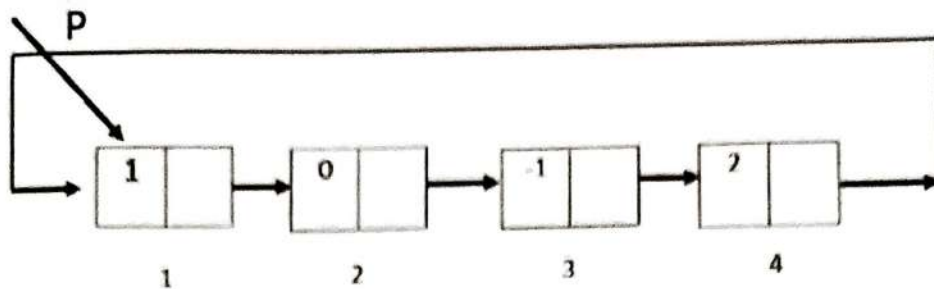
(CO3:8 Marks)

III.A.2 You are asked to **take a class for first-year computer science students** on the topic "**Applications of Linked Lists**." Your goal is to help them understand **where and why linked lists are used in real-world programming**. **Identify and explain different applications** of linked lists in software systems or computing environments. (CO3:7 Marks)

OR

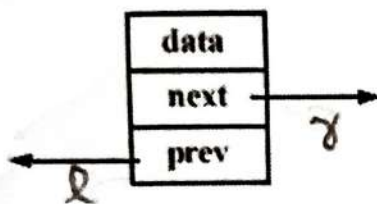
III.B.1 Consider the following circular linked list. Explain the algorithm and illustrate the insertion of a new node with item 79 at position 5, which

will become as the last node of the list. With the given list, 'P' is assumed to be the pointer to the first node.

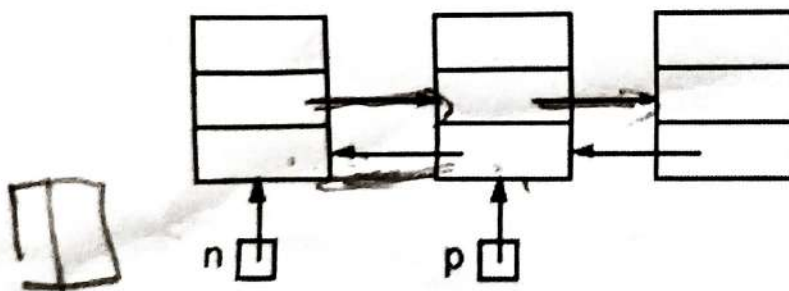


(CO3:10 Marks)

III.B.2 Nodes for a doubly linked list are defined to have the following structure.



The next instance variable stores a reference to the next node in the list, and the prev instance variable refers to the previous node in the list. Below is a list of three of these nodes, along with two reference variables, p and n, that refer to specific nodes in the list as shown.



Which of the following expressions does not refer to the third node in the list and justify your answer.

- (i). $p \rightarrow \text{next}$ (ii). $n \rightarrow \text{next} \rightarrow \text{next}$ (iii). $p \rightarrow \text{prev} \rightarrow \text{next}$ (iv). $p \rightarrow \text{prev} \rightarrow \text{next} \rightarrow \text{next}$

(CO3:5 Marks)

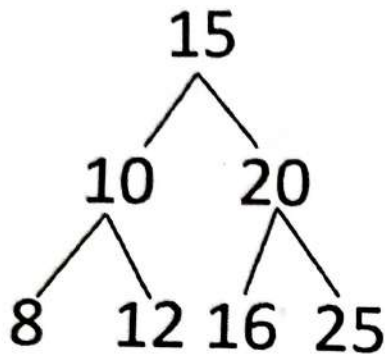
BUNCH IV

IV.A Build and illustrate the Binary Search Tree (BST) created by adding the numbers {12, 5, 18, 2, 9, 15, 19, 13, 17} in the specified sequence. Then, search for the number 10 within this BST. State if the number 10 is present or absent and detail the process followed during the search.

(CO4:15 Marks)

OR

IV.B.1 Consider the following Binary search tree and demonstrate the insertion of the item 24 in it.



(CO4:10 Marks)

IV.B.2 You are a software engineer working on developing a library for manipulating binary trees. Your colleague, Jane, asserts that "All binary trees are not binary search trees". She argues that while some binary trees can be binary search trees; not all binary trees adhere to the binary search tree property. As part of your team's ongoing discussions, you are tasked with investigating Jane's assertion and providing a thorough justification for your stance. Present your findings with clear examples and explanations to support your argument.

(CO4:5 Marks)

CO1	Illustrate the fundamental concepts of Data Structures, the Representation of single and two-dimensional arrays and the implementation of various operations on them
CO2	Analyse the representation of stacks and queues using arrays, operations on them and application of these data structures in problem-solving.
CO3	Demonstrate the ability to implement and manipulate various types of linked lists (singly, doubly, and circular)
CO4	Demonstrate the basic concepts of Graphs and Trees and the operations on Binary search trees.
CO5	Implement Data Structures using C++ programming